

Mind the (DH) Gap! A Contrast in Risky Choices Between Reasoning and Conversational LLMs

注意 (DH) 差距！推理与对话型大语言模型在风险选择上的对比

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Abstract

摘要

The use of large language models either as decision support systems, or in agentic workflows, is rapidly transforming the digital ecosystem. However, the understanding of LLM decision-making under uncertainty remains limited. We study LLM risky choices along two dimensions: (1) prospect representation (based on an explicit representation or outcome history) and (2) decision rationale (explanation). Our study, which involves 20 frontier and open LLMs, is complemented by a matched human subjects experiment, which provides one reference point, while an expected payoff maximizing rational agent model provides another. We find that LLMs cluster into two categories: reasoning models (RMs) and conversational models (CMs). RMs tend towards rational behavior, are insensitive to the order of prospects, gain/loss framing, and explanations, and behave similarly whether prospects are explicit or presented via a history of outcomes. CMs are significantly less rational, slightly more human-like, sensitive to prospect ordering, framing, and explanation, and exhibit a large description-history gap. Paired comparisons of open LLMs suggest that a key factor differentiating RMs and CMs is training for mathematical reasoning.

使用大型语言模型作为决策支持系统或在代理工作流中的应用，正在迅速改变数字生态系统。然而，对在不确定性条件下 LLM 决策的理解仍然有限。我们从两个维度研究 LLM 的风险选择：(1) 前景表示（基于显式表示或结果历史）和 (2) 决策理由（解释）。我们的研究涉及 20 个前沿和开放的 LLM，并辅以匹配的人类受试者实验，提供了其中一个参考点，而预期收益最大化理性代理模型提供了另一个。我们发现，LLM 可以分为两类：推理模型（RMs）和对话模型（CMs）。RMs 倾向于理性行为，对前景顺序、收益/损失框架和解释不敏感，并且无论前景是显式还是通过结果历史呈现，其行为都相似。CMs 则明显不够理性，略微更像人类，对前景顺序、框架和解释敏感，并表现出较大的描述-历史差异。对开放 LLM 的配对比较表明，区分 RMs 和 CMs 的一个关键因素是数学推理的训练。

1 Introduction

1 引言

Large language models (LLMs) are increasingly used in decision support (Benary et al., 2023; Vrdoljak et al., 2025) as well as agentic workflows that invoke tools and execute multi-step plans (Moncada-Ramirez et al., 2025; Webb et al., 2025). Consequently, LLMs are becoming significant economic decision-makers, with uncertainty a central consideration. Traditional computational paradigms for decision-making under uncertainty choose an option that maximizes expected utility or payoff (Parkes and Wellman, 2015). On the other hand, humans are known to systematically

deviate

大型语言模型 (LLMs) 越来越多地用于决策支持 (Benary 等, 2023; Vrdoljak 等, 2025), 以及调用工具并执行多步骤计划的自主工作流程 (Moncada-Ramirez 等, 2025; Webb 等, 2025)。因此, LLMs 正在成为重要的经济决策者, 不确定性成为核心考虑因素。传统的不确定性决策计算范式选择能够最大化预期效用或收益的选项 (Parkes 和 Wellman, 2015)。另一方面, 人类被发现会系统性地偏离

from such behavior, and a host of mathematical decision models has been proposed to capture human risky choice (Connolly and Butler, 2006; Kahneman and Tversky, 1979; Peterson et al., 2021).

从这种行为出发, 已经提出了一系列数学决策模型来捕捉人类的风险选择 (Connolly 和 Butler, 2006; Kahneman 和 Tversky, 1979; Peterson 等, 2021)。

However, LLMs are neither explicitly designed to maximize expected payoff, nor necessarily behave like humans. Rather, their competence is an emergent property of scale and training paradigms including next-word prediction, instruction fine-tuning, human preference alignment, and mathematical reasoning (Du et al., 2024; Guo et al., 2025; Ouyang et al., 2022). The net effect of this soup of training methods on LLM economic decision-making under uncertainty remains unclear. In response, a literature has begun to emerge that aims to investigate LLM decision-making, particularly under uncertainty (Binz and Schulz, 2023; Coda-Forno et al., 2024; Jia et al., 2024; Ross et al., 2024; Payne, 2025; Wang et al., 2025). However, there remain conflicting accounts, for example, of the extent LLMs are rational or align with prospect theory and human behavior (Chen et al., 2023; Horton, 2023; Jia et al., 2024; Ross et al., 2024; Wang et al., 2025; Payne, 2025). More fundamentally, given the broad variety of training paradigms, goals, and architectures, it is unclear whether a straightforward account of LLM economic decision making is even possible.

然而, 大型语言模型 (LLMs) 并非专门设计以最大化预期收益, 也不一定像人类一样行为。相反, 它们的能力是规模和训练范式的涌现属性, 包括下一个词预测、指令微调、人类偏好对齐以及数学推理 (Du 等, 2024; Guo 等, 2025; Ouyang 等, 2022)。这种多种训练方法的综合对 LLM 在不确定性下的经济决策的净效应仍不清楚。对此, 已有文献开始出现, 旨在研究 LLM 的决策, 特别是在不确定性下的决策 (Binz 和 Schulz, 2023; Coda-Forno 等, 2024; Jia 等, 2024; Ross 等, 2024; Payne, 2025; Wang 等, 2025)。然而, 仍存在相互矛盾的观点, 例如关于 LLM 是否理性或符合前景理论和人类行为的程度 (Chen 等, 2023; Horton, 2023; Jia 等, 2024; Ross 等, 2024; Wang 等, 2025; Payne, 2025)。更根本的是, 鉴于训练范式、目标和架构的广泛多样性, 甚至很难确定是否能够对 LLM 的经济决策做出简单的解释。

We study the comparative behavior of LLMs in a minimal, controlled two-option risky-choice benchmark, focusing on two underexplored dimensions: 1) how the prospects (uncertain options) are represented and 2) the impact of requesting LLMs to explain their decisions. The former is motivated by the description–experience (D–E) gap in human risky choice (Hertwig et al., 2004; Hertwig and Erev, 2009; Wulff et al., 2018), which documents that people can choose differently when the same underlying prospects are learned through experience rather than read as explicit descriptions. The latter is motivated by evidence that verbalizing rea-

我们研究了在最小化、可控的两选项风险选择基准中, 大语言模型 (LLMs) 的比较行为, 重点关注两个尚未充分探索的维度: 1) 前景 (不确定选项) 是如何被表示的, 以及 2) 要求大语言模型解释其决策的影响。前者受到人类风险选择中描述–经验 (D–E) 差距的启发 (Hertwig 等, 2004; Hertwig 和 Erev, 2009; Wulff 等, 2018), 该研究指出, 当相同的前景是通过经验学习而不是通过明确描述阅读时, 人们可能会做出不同的选择。后者则受到证据的启发, 即口头表达理由可以影响决策过程。

sons can change human judgments and preferences (Festinger, 1962; Shafir et al., 1993; Wilson and Schooler, 1991), and by the practical expectation that AI systems should be explainable (Ferrario and Loi, 2022; Zhao et al., 2024).

儿子可以改变人类的判断和偏好 (Festinger, 1962; Shafir 等人, 1993; Wilson 和 Schooler, 1991), 并且基于实践期望, 人工智能系统应该具有可解释性 (Ferrario 和 Loi, 2022; Zhao 等人, 2024)。

We consider 20 frontier and open LLMs, along with human subjects' choices over the same prospects and treatments. LLMs are analyzed vis-a-vis two references: 1) the human subjects pool, and 2) the idealized economicus rational agent. Within the scope of our experiments, our key findings are:

我们考虑了 20 个前沿且开放的大型语言模型 (LLMs)，并结合了人类受试者在相同前景和治疗方案下的选择。这些语言模型是相对于两个基准进行分析的：1) 人类受试者群体，以及 2) 理想化的经济人理性代理。在我们实验的范围内，我们的主要发现如下：

1. LLMs cluster into two behavioral categories: reasoning models (RMs) and conversational models (CMs). RMs are similar to the economicus while CMs are distinct from both the economicus and the human subjects pool.
2. All models exhibit a description-history (DH) gap, an analogue of the decision-experience (DE) gap in which agents only see a history of past outcomes. This gap is considerable for CMs, but modest for RMs.
3. Explanations impact decisions for all models, though the impact is greater for CMs. Surprisingly, they are typically more aligned with economicus under brief explanations than under no explanation or mathematical explanations.
4. Paired analysis of open models suggests that fine-tuning for mathematical reasoning is a key differentiator between RMs and CMs. Other parts of the training pipeline, however, appear to have limited impact.
5. 大型语言模型 (LLMs) 可以分为两类行为：推理模型 (RMs) 和对话模型 (CMs)。推理模型 (RMs) 与经济人 (economicus) 相似，而对话模型 (CMs) 则与经济人和人类受试者群体都不同。
6. 所有模型都表现出描述历史 (DH) 差距，这是决策经验 (DE) 差距的类比，其中代理只能看到过去结果的历史。这种差距对于对话模型 (CMs) 来说相当显著，但对于推理模型 (RMs) 来说则较为有限。
7. 解释对所有模型的决策都有影响，尽管对对话模型 (CMs) 的影响更大。令人惊讶的是，它们通常在简短解释下比在无解释或数学解释下更符合经济人 (economicus) 的预期。
8. 对开放模型的配对分析表明，针对数学推理的微调是推理模型 (RMs) 和对话模型 (CMs) 之间的一个关键区别。然而，训练流程中的其他部分似乎影响有限。

Related Work. LLMs are increasingly studied as objects of behavioral analysis (Binz and Schulz, 2023; Coda-Forno et al., 2024; Dillion et al., 2023; Hagendorff et al., 2023; Hayes et al., 2024; Horton, 2023; Ivanova, 2023). Jia et al. (2024) analyze LLM decisions by eliciting choices and fitting prospect-theoretic models; subsequent work extends this by examining epistemic markers (Wang et al., 2025), application contexts (Payne, 2025), and persona matching (Liu et al., 2025). Horowitz and Plonsky (2025) study a description-experience gap in LLMs through repeated choice, unlike the passive sampling used in our setting. Apart from the bounded rationality focus, Chen et al. (2023) instead focuses on rationality (consistency with utility maximization) of GPTs directly (see also (Jiang et al., 2025) for a survey) and Mazeika et al. (2025) elicits the inherent value functions of various LLMs, while Coda-Forno et al. (2024) provide a benchmarking suite for LLM decision making under risk. We provide a broader discussion of related research in Appendix A. In general, existing work tends to focus narrowly on a single theory, relies on limited model coverage, overlooks the effects of choice representation and output format, or lacks matched human comparisons. Our work aims to address some of these limitations.

相关工作。 大语言模型 (LLMs) 正越来越多地作为行为分析的对象被研究 (Binz 和 Schulz, 2023;

Coda-Forno 等, 2024; Dillion 等, 2023; Hagendorff 等, 2023; Hayes 等, 2024; Horton, 2023; Ivanova, 2023)。Jia 等 (2024) 通过获取选择并拟合前景理论模型来分析大语言模型的决策; 后续的研究扩展了这一方法, 通过考察知识标记 (Wang 等, 2025)、应用场景 (Payne, 2025) 以及人格匹配 (Liu 等, 2025) 来进行分析。Horowitz 和 Plonsky (2025) 通过重复选择研究大语言模型中的描述-经验差距, 这与我们在研究中使用的被动抽样方法不同。除了对有限理性的关注外, Chen 等 (2023) 则直接关注 GPTs 的理性 (与效用最大化的一致性) (参见 Jiang 等, 2025 的综述), Mazeika 等 (2025) 则获取了各种大语言模型的内在价值函数, 而 Coda-Forno 等 (2024) 则提供了一套用于在风险条件下评估大语言模型决策的基准测试工具。我们在附录 A 中对相关研究进行了更广泛的讨论。总体而言, 现有研究往往局限于单一理论, 模型覆盖范围有限, 忽视了选择表示和输出格式的影响, 或者缺乏与人类的匹配比较。我们的工作旨在解决这些限制中的一些问题。

2 Study Design

2 研究设计

We investigate how large language models (LLMs) make decisions when faced with uncertainty. To do this comprehensively, we examine 20 different LLMs (detailed in the Supplement, Appendix B.1). Our selection covers two important goals: first, we include widely used frontier models; second, we use open-weight models of varying sizes and training stages to enable controlled comparisons. We query all LLMs through a unified interface using a minimal instruction template (see Supplement, Appendix B.3 for our prompts and an additional prompt sensitivity analysis). To measure behavior, we compare LLM responses against two references: human subject behavior and *economicus*, a risk-neutral expected payoff maximizer.

我们研究大型语言模型 (LLMs) 在面对不确定性时如何做出决策。为了全面地进行这项研究, 我们考察了 20 个不同的 LLMs (详细信息见补充材料, 附录 B.1)。我们的选择涵盖了两个重要的目标: 第一, 我们包含了广泛使用的前沿模型; 第二, 我们使用了不同规模和训练阶段的开源权重模型, 以便进行可控的比较。我们通过一个统一的接口, 使用一个最小化的指令模板查询所有 LLMs (有关我们的提示以及额外的提示敏感性分析, 请参见补充材料, 附录 B.3)。为了衡量行为, 我们将 LLM 的响应与两个参考标准进行比较: 人类受试者的行为和经济人 (*economicus*), 即一个风险中性且最大化预期收益的模型。

2.1 Choices Among Prospects

2.1 对前景的选择

Our study uses three base prospect pairs as a structured testbed. These pairs span different outcome scales and are intentionally similar to (but not duplicates of) classic behavioral-economics stimuli (Kahneman and Tversky, 2013; Peterson et al., 2021). This design reduces the risk of LLM memorization. See Appendix B.4 for details.

我们的研究使用三个基础前景对作为结构化的测试平台。这些对涵盖不同的结果尺度, 并有意与经典行为经济学刺激相似 (但并非重复), 如 (Kahneman 和 Tversky, 2013; Peterson 等, 2021)。这种设计降低了大语言模型记忆的风险。详情请参见附录 B.4。

Description-History Gap. Human decision-making differs depending on how information is presented. Research in the description-experience (DE) gap shows that people decide differently when given explicit information versus when they learn through experience. We adapt this insight to study LLMs. Rather than having LLMs interact dynamically with an environment (which would introduce confounds), we present prospects in two ways:

描述-历史差距。 人类的决策会根据信息呈现方式的不同而有所差异。关于描述-经验 (DE) 差距的研究表明, 当人们获得明确信息时, 他们的决策方式与通过经验学习时有所不同。我们将这一见解应用于研究大型语言模型 (LLMs)。我们不是让 LLMs 与环境动态交互 (这会引入混淆因素), 而是以两种方式呈现前景:

- **Explicit Prospects.** Each prospect specifies exact payoff-probability pairs. For example: “70%

chance of \$100, 30% chance of \$0.”

- **Implicit Prospects.** We present prospects as simulated histories – sequences of payoffs that would result from repeated selection. For instance, a simulated history might show 15 instances of \$100 and 5 instances of \$0 from 20 trials, representing the same underlying distribution.

- **明确前景。**每个前景都指定了具体的收益-概率对。例如：“有 70% 的机会获得 \$100, 30% chance of \$0。”

- **暗示前景。**我们将前景呈现为模拟的历史记录——由重复选择所产生的收益序列。例如，一个模拟的历史记录可能显示在 20 次试验中有 15 次出现 \$100 and 5 instances of \$0，这代表了相同的潜在分布。

We call the behavioral difference between these presentations the description–history (DH) gap. We test with simulated histories of 20 and 100 payoffs, with results aggregated across sample sizes (see Appendix C.3 for breakdowns by size).

我们称这些展示之间的行为差异为描述–历史 (DH) 差距。我们使用 20 和 100 个收益的模拟历史进行测试，结果在不同样本量下进行汇总（有关按样本量细分的详细信息，请参见附录 C.3）。

Decisions and Explanations. Our second key consideration involves the impact that requesting LLMs to provide a reason (explanation) has on its decisions. We implement this using three prompt styles that request either 1) no explanation (output the choice only), 2) a one-sentence justification (short explanation), or 3) a brief mathematical or reasoning-style justification (math explanation). See the Supplement (Appendix B.3) for details.

决策与解释。我们的第二个关键考虑因素涉及请求大型语言模型 (LLMs) 提供理由（解释）对其决策的影响。我们通过三种提示风格来实现这一点：1) 不提供解释（仅输出选择），2) 提供一个句子的说明（简短解释），或 3) 提供一个简短的数学或推理风格的解释（数学解释）。详细信息请参见补充材料（附录 B.3）。

2.2 Human Subjects Experiments

2.2 人体受试者实验

To compare model behavior with human decision-making, we collected responses from 360 U.S.-based participants via Prolific. Each participant selects among the same set of prospect pairs as LLMs and is compensated at an average rate of \$24/hour. In analysis, we treat humans as a single population distribution over choices. This study was approved by the institutional IRB. We also conducted an initial attention check by excluding participants who spent an average of less than 8 seconds per question. As this exclusion did not lead to significant changes in the results, we report the analyses in the main text including all participants.

为了将模型的行为与人类的决策进行比较，我们通过 Prolific 收集了 360 名美国参与者的意见。每位参与者在与 LLMs 相同的前景对集合中进行选择，并以平均每小时 24 美元的费率获得报酬。在分析中，我们将人类视为在选择上的一组总体分布。本研究已获得机构 IRB 的批准。我们还通过排除平均每个问题花费少于 8 秒的参与者来进行初步的注意力检查。由于这种排除并未导致结果产生显著变化，我们在正文中报告的分析包括所有参与者。

2.3 Interpretable Models of Behavior

2.3 可解释的行为模型

To complement our analysis of the raw behavioral data, we make use of interpretable models of (LLM or human) behavior to obtain deeper insights into behavior and effective risk preferences. We consider two parameterizations of prospect theory, both with four free parameters (presented in full in Appendix B.2.1). The first is a standard formulation with parameters σ (risk preference), λ (loss aversion), γ (probability weighting), and β (decisiveness). However, because λ and β are not jointly identifiable in pure-loss prospects, we also consider an alternative specification as

our primary model (a special case of the generalized power value model (Peterson et al., 2021)), replacing λ and β with β_{gain} and β_{loss} for the gain and loss frames, respectively, while retaining σ and γ . In these models:

为了补充我们对原始行为数据的分析，我们利用可解释的行为模型（LLM 或人类）来获得对行为和有效风险偏好的更深入理解。我们考虑了前景理论的两种参数化形式，两者均包含四个自由参数（完整参数见附录 B.2.1）。第一种是标准形式，参数为 σ （风险偏好）、 λ （损失厌恶）、 γ （概率加权）和 β （决断性）。然而，由于在纯损失前景中 λ 和 β 无法联合识别，我们还考虑了一种替代的参数化形式作为主要模型（广义幂值模型的一个特例（Peterson 等，2021）），在收益和损失框架中分别用 β_{gain} 和 β_{loss} 替代 λ 和 β ，同时保留 σ 和 γ 。在这些模型中：

- Higher σ implies risk seeking, $\sigma \rightarrow 0$ entails risk aversion, while $\sigma = 1$ is risk neutral.
- Higher γ implies underweighting of small probabilities, $\gamma \rightarrow 0$ indicates overweighting of these,
- 更高的 σ 表示风险偏好， $\sigma \rightarrow 0$ 表示风险规避，而 $\sigma = 1$ 表示风险中性。
- 更高的 γ 表示对小概率事件的低估， $\gamma \rightarrow 0$ 表示对这些事件的高估，

and $\gamma = 1$ means no probability distortion.

and $\gamma = 1$ means no probability distortion.

- Higher β_{gain} (and β_{loss}) indicates greater determinism in the gain (and loss) domain, while $\beta \rightarrow 0$ implies essentially uniformly random choice. The ratio $\beta_{loss}/\beta_{gain}$ serves as a proxy for loss aversion: a ratio greater than 1 indicates higher sensitivity to utility differences in losses than in gains.

- 更高的 β_{gain} (和 β_{loss}) 表明在收益 (和损失) 领域具有更大的确定性，而 $\beta \rightarrow 0$ 则意味着基本上是随机选择。比率 $\beta_{loss}/\beta_{gain}$ 可作为损失厌恶的代理指标：比率大于 1 表示对损失的效用差异比对收益的效用差异更为敏感。

To avoid pathologies, we bound all parameters in $[0.01, 1000]$. An economicus then roughly corresponds to $\sigma = \gamma = 1$ and $\beta_{gain} = \beta_{loss} = 1000$.

为了避免出现病态情况，我们将所有参数限制在 $[0.01, 1000]$ 范围内。此时，一个经济人大致对应于 $\sigma = \gamma = 1$ 和 $\beta_{gain} = \beta_{loss} = 1000$ 。

2.4 Quantifying Behavior

2.4 行为量化

Querying and parsing. For each model and instantiated condition, we estimate a response distribution through repeated querying of 10 times under identical inputs. We use temperature $T = 1$ and a maximum generation length of 1024 tokens. We found that increasing the number of samples to 50 has minimal qualitative impact on the results, while being significantly more costly. If an LLM output does not contain a valid prospect choice, we retain the raw output and mark the trial as invalid. Choice rates are computed over valid trials.

查询和解析。对于每个模型和实例化的条件，我们通过在相同输入下重复查询 10 次来估计响应分布。我们使用温度 $T = 1$ 和最大生成长度为 1024 个标记。我们发现将样本数量增加到 50 对结果的定性影响很小，但成本显著增加。如果 LLM 的输出不包含有效的前景选择，我们将保留原始输出并标记该试验为无效。选择率是在有效试验上计算的。

Model Similarity and Goodness of Fit. Let x refer to a pair of prospects, as well as a particular way of presenting these, which includes: prospect order, framing (loss vs. gain), nature of explanation requested, and prospect presentation (explicit vs. implicit, and the length of the history for the latter). X denotes the set of all such contexts.

模型相似性与拟合度。设 x 表示一对前景，以及呈现这些前景的特定方式，包括：前景顺序、框架（损失与收益）、所请求的解释性质，以及前景呈现方式（显性与隐性，后者的历史长度）。 X 表示所有此类情境的集合。

For a given context x and a model m (e.g., LLM, or a parametric model that we fit to data collected from LLM or human choices), we use $p_m(x) \in [0, 1]$ to denote the fraction of times (probability) that m selects the reference prospect (whichever we choose it to be) in this context. For the economicus $p_e(x)$ is deterministic, while $p_h(x)$ denotes the fraction of human subjects who selected the reference prospect.

对于给定的上下文 x 和一个模型 m （例如，LLM，或我们拟合到从 LLM 或人类选择中收集的数据的参数化模型），我们使用 $p_m(x) \in [0, 1]$ 表示 m 在该上下文中选择参考前景（无论我们选择它为何种）的次数（概率）。对于经济人（economicus）， $p_e(x)$ 是确定性的，而 $p_h(x)$ 表示选择参考前景的人类受试者的比例。

For a pair of models m and m' (which can also refer to h or e), we can measure how well m predicts (or fits) the behavior of m' in two ways. The first is mean-squared error (MSE):

对于一对模型 m 和 m' （也可以指 h 或 e ），我们可以用两种方式来衡量 m 对 m' 行为的预测（或拟合）效果。第一种是均方误差（MSE）：

$$\text{MSE}(p_m, p_{m'}) = \frac{1}{|X|} \sum_{x \in X} (p_m(x) - p_{m'}(x))^2.$$

However, when MSE is non-zero, it may fail to capture an important property that decision patterns “go together”, i.e., the extent to which an increase or decrease in p_m across different contexts is accompanied by a similar change in $p_{m'}$.

然而，当均方误差（MSE）不为零时，它可能无法捕捉到一个重要的属性，即决策模式“相互关联”，也就是说，在不同情境下， p_m 的增加或减少伴随着 $p_{m'}$ 的相似变化程度。

To capture this, we treat $p_m(x)$ as a random variable with context x viewed as the associated outcome. We can then measure similarity between a pair of models m and m' using Pearson correlation between responses over all contexts $p_m(X)$ and $p_{m'}(X)$, $\text{Cor}(p_m(X), p_{m'}(X))$. If m is an interpretable parametric model fit to data, we evaluate both MSE and correlation on a held out (test) dataset that is distinct from the data on which the parameters of m were fit. Since the ability to capture the trends of behavior across contexts x is essential to our analysis, we use correlation as the primary measure of both (a) similarity of pairs of models (e.g., LLMs), and (b) goodness of fit of a parametric model. Nevertheless, we observe that both MSE and correlation typically lead to the same goodness-of-fit conclusions.

为了捕捉这一点，我们将 $p_m(x)$ 视为一个随机变量，将上下文 x 视为相关的结果。然后，我们可以使用所有上下文 $p_m(X)$ 和 $p_{m'}(X)$ 以及 $\text{Cor}(p_m(X), p_{m'}(X))$ 上的响应之间的皮尔逊相关系数来衡量一对模型 m 和 m' 之间的相似性。如果 m 是一个可解释的参数化模型，并且是基于数据拟合的，我们将在一个与用于拟合 m 参数的数据不同的保留（测试）数据集上评估均方误差（MSE）和相关性。由于捕捉在上下文 x 上行为趋势的能力对我们的分析至关重要，因此我们将相关性作为衡量以下两方面的主要指标：(a) 模型对（例如大型语言模型）之间的相似性，以及 (b) 参数化模型的拟合优度。然而，我们观察到，均方误差和相关性通常会导致相同的拟合优度结论。

Decisiveness and Consistency. In addition to model similarity and goodness of fit, two other behavioral traits we explore are decisiveness, which captures the entropy in the decisions, and con-

sistency, of which we consider three forms. The first is order consistency, capturing the degree to which decisions tend to differ as a function of the order in which the prospects are presented. The second is prompt consistency, which measures the impact of explanation types on the nature of decisions. Finally, we consider frame consistency, which captures the impact of the gain vs. loss framing of otherwise identical prospects on the decision.

果断性与一致性。除了模型相似性和拟合度之外，我们还探讨另外两个行为特征：果断性，它捕捉了决策中的熵；以及一致性，我们考虑了三种形式。第一种是顺序一致性，它衡量决策在呈现前景顺序变化时的差异程度。第二种是即时一致性，它衡量解释类型对决策本质的影响。最后，我们考虑框架一致性，它捕捉了在其他条件完全相同的情况下，收益与损失框架对决策的影响。

Formally, decisiveness of a model m (or a human h) is defined as $\frac{1}{|X|} \sum_{x \in X} \max\{p_m(x), 1 - p_m(x)\}$. For consistency measures, let $S(x)$ define a collection of variations from the base context x that we consider. For example, in the case of ordering, $S(x)$ is comprised of before and after a given prospect is switched. Order and prompt consistency is then defined as $1 - \frac{1}{|X|} \sum_{x \in X} |\max_{x' \in S(x)} p_m(x') - \min_{x' \in S(x)} p_m(x')|$. Frame consistency is defined as $1 - \frac{1}{|X|} \sum_{x \in X} |p_m(\text{gain}) - (1 - p_m(\text{loss}))|$.

正式来说，模型 m (或人类 h) 的果断性被定义为 $\frac{1}{|X|} \sum_{x \in X} \max\{p_m(x), 1 - p_m(x)\}$ 。对于一致性度量，设 $S(x)$ 定义一个从基础上下文 x 考虑的变体集合。例如，在订购的情况下， $S(x)$ 包括在给定前景切换前后的部分。顺序和提示一致性则被定义为 $1 - \frac{1}{|X|} \sum_{x \in X} |\max_{x' \in S(x)} p_m(x') - \min_{x' \in S(x)} p_m(x')|$ 。框架一致性被定义为 $1 - \frac{1}{|X|} \sum_{x \in X} |p_m(\text{gain}) - (1 - p_m(\text{loss}))|$ 。

3 Behavior Patterns and Convergence

3 行为模式与收敛性

We begin our analysis by considering aggregate behavior for each LLM, as well as that of human and economicus references. In Figure 1 we present pairwise correlations between pairs of frontier LLM models, as well as humans and economicus.

我们首先通过考虑每个大语言模型 (LLM) 的整体行为，以及人类和经济人 (economicus) 的行为来开始我们的分析。在图 1 中，我们展示了前沿大语言模型之间、以及人类和经济人之间的成对相关性。

An immediate observation from the figure is the emergence of two LLM clusters: one that includes the (frontier) reasoning models GPT-5.1,

从图中可以立即观察到出现两个 LLM 集群：一个包括 (前沿) 推理模型 GPT-5.1,

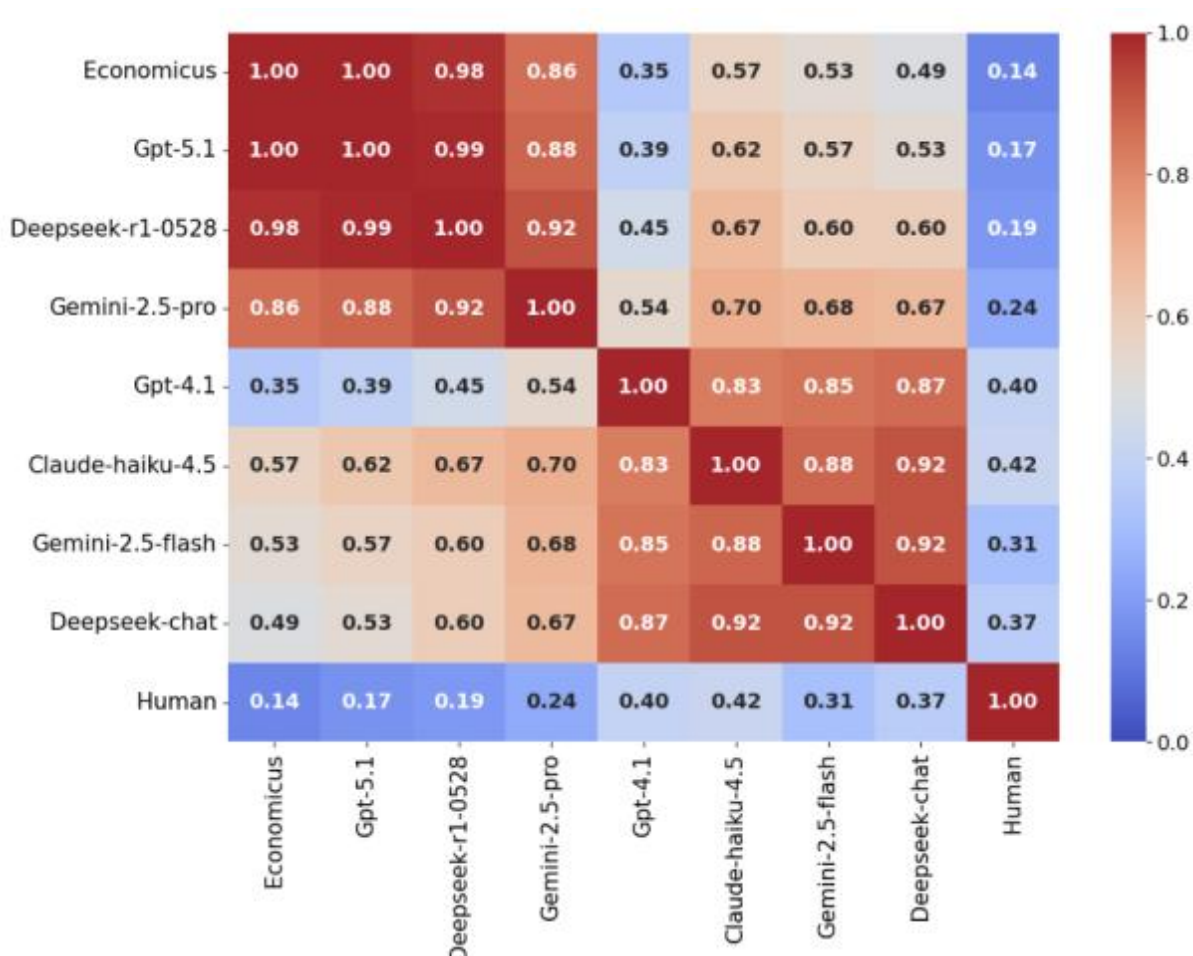


Figure 1: images/828dd143e03369525570eac5a730940937978e01cf91231cd05613b2083e11e9.jpg

Figure 1: Correlation matrix involving (1) frontier LLMs (GPT-4.1 and 5.1, Gemini-2.5-Flash and Pro, DeepSeek-R1 and Chat, and Claude-Haiku-4.5), (2) economicus, and (3) human responses.

图 1: 涉及 (1) 前沿大语言模型 (GPT-4.1 和 5.1、Gemini-2.5-Flash 和 Pro、DeepSeek-R1 和 Chat、以及 Claude-Haiku-4.5)、(2) 经济人, 以及 (3) 人类响应的相关性矩阵。

DeepSeek-R1, and Gemini-2.5-Pro, and the other including their conversational counterparts (GPT-4.1, DeepSeek-Chat, and Gemini-2.5-Flash), as well as Claude-4.5-Haiku. In particular, correlations among pairs of models within each cluster are > 0.8 , and in most cases ≈ 0.9 or higher. In contrast, correlations among pairs of models across clusters tend to be considerably lower—typically ≈ 0.6 or below. This appears to be an example of convergence of frontier LLMs (Smith et al., 2025; Zhou et al., 2025; Mazeika et al., 2025), albeit across two distinct lines that reflect their target use. Henceforth, we thus distinguish between reasoning models (RMs) and conversational models (CM) in terms of their respective decision-making behavior under uncertainty. This split into RMs and CMs is also observed in open models; see the full heatmap in the Supplement (Appendix C.1).

DeepSeek-R1, Gemini-2.5-Pro, 以及其他包括其对话版本 (GPT-4.1, DeepSeek-Chat 和 Gemini-2.5-Flash), 以及 Claude-4.5-Haiku。特别是, 在每个集群内的模型对之间的相关性大于 0.8, 并且在大多数情况下 $\approx$ 高于 0.9。相比之下, 跨集群的模型对之间的相关性通常要低得多——通常 $\approx$ 低于 0.6。这似乎是一个前沿大语言模型收敛的例子 (Smith 等, 2025; Zhou 等, 2025; Mazeika 等, 2025), 尽管它们反映了各自的目标用途, 沿着两条不同的路线。因此, 我们根据它们在不确定性下的决策行为, 将模型区分为推理模型 (RMs) 和对话模型 (CMs)。这种 RMs 和 CMs 的划分在开源模型中也观察到了; 参见补充材料中的完整热力图 (附录 C.1)。

Our next finding is that neither RMs nor CMs are particularly similar to human behavior. This observation is reflected in the remarkably low pairwise correlation in Figure 1 between each frontier LLM and human: the highest is with the Claude Haiku model at only 0.42. However, we do observe an intuitive pattern: RMs are considerably more different from human behavior than the CMs.

我们的下一个发现是，无论是 RMs 还是 CMs，都不特别类似于人类的行为。这一观察在图 1 中得到了体现：每个前沿 LLM 与人类之间的成对相关性出奇地低，最高的仅与 Claude Haiku 模型达到 0.42。然而，我们确实观察到了一个直观的模式：RMs 与人类行为的差异要比 CMs 大得多。

On the other hand, RMs are quite similar to the economicus reference, with correlations ranging from 0.86 to 1. In contrast, CMs exhibit low similarity to economicus, with correlations below 0.5. We also note that human behavior is essentially uncorrelated with economicus, with correlation only 0.14. This is likely a consequence of selecting prospects that most emphasize the deviation of hu-

另一方面，RM 与经济人参考模型非常相似，相关性范围在 0.86 到 1 之间。相比之下，CM 与经济人之间的相似性较低，相关性低于 0.5。我们还注意到，人类行为与经济人几乎无相关性，相关性仅为 0.14。这很可能是由于选择了最强调人类偏离的前景所致。

man behavior from expected utility maximization.

人的行为来自预期效用最大化。

To confirm that these structural differences are robust to sampling noise, we conducted an empirical bootstrap analysis: for each model and prospect pair, we resampled 1,000 synthetic datasets from a Binomial distribution parameterized by the observed choice frequency, and recomputed all behavioral metrics. The RM-CM clustering, as well as the deviation of human behavior from both, hold with $p < 0.001$ across all comparisons.

为了确认这些结构差异对抽样噪声具有鲁棒性，我们进行了实证的自助法分析：对于每个模型和前景对，我们从以观察到的选择频率参数化的二项式分布中重新抽样 1,000 个合成数据集，并重新计算所有行为指标。RM-CM 聚类，以及人类行为与这两者的偏差，在所有比较中均以 $p < 0.001$ 保持显著。

To simplify presentation, in Figure 2, we introduce the HE representation, plotting each model on a two-dimensional space anchored by human and economicus correlations (note humans have a non-zero correlation to the economicus). This visualization provides a compact, interpretable framework for understanding LLM behavior relative to two behavioral extremes. The RM and CM clusters emerge here as well, with open models mapping predictably to each. For example, Qwen2.5-7B-Instruct is a CM model, while its mathematical reasoning variant is clearly an RM model (being far more economicus-like and somewhat less human-like than the instruct version). We see the same pattern with Qwen3-30B as well as with Olmo-3-7B (although here, the “Think” variant is slightly more human-like). In general, we observe that RMs tend to be far more economicus-like, and (usually) slightly less human-like compared to CMs.

为简化展示，在图 2 中，我们引入了 HE 表示法，将每个模型绘制在由人类和 economicus 相关性锚定的二维空间中（请注意，人类与 economicus 的相关性不为零）。这种可视化方法提供了一个紧凑且易于理解的框架，用于理解大型语言模型（LLM）的行为相对于两种行为极端情况的表现。在这里，RM 和 CM 聚类也显现出来，开放模型会以可预测的方式映射到各自的类别。例如，Qwen2.5-7B-Instruct 是一个 CM 模型，而其数学推理变体则明显是一个 RM 模型（因为它比指令版本更像 economicus，且在某种程度上更不像人类）。我们同样在 Qwen3-30B 以及 Olmo-3-7B 中观察到了相同的模式（尽管在这里，“Think”变体略微更像人类）。总体而言，我们观察到 RM 通常比 CM 更像 economicus，并且（通常）比 CM 稍少像人类。

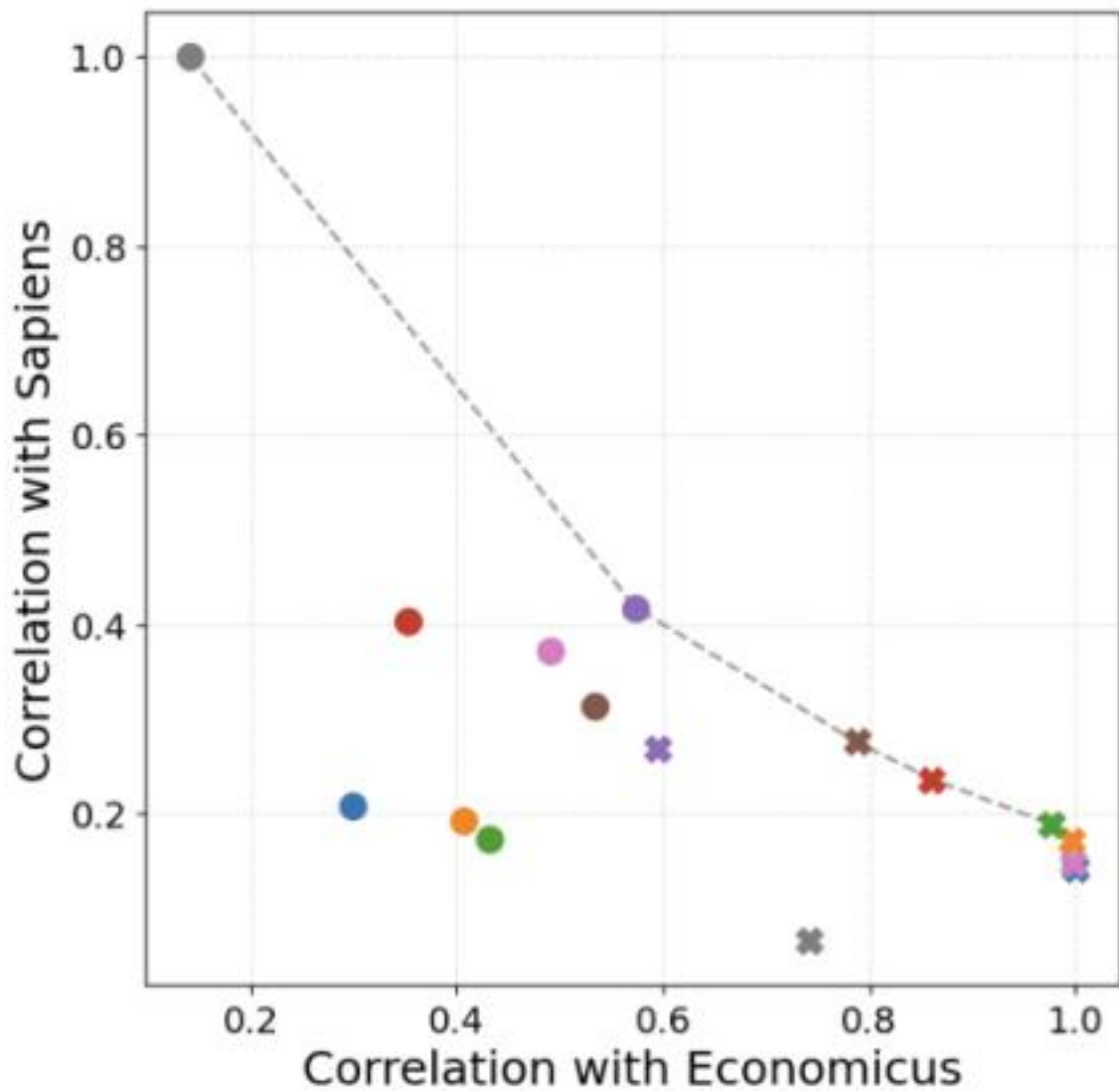


Figure 2: images/8b7596b5f50aaf6fe691b14a747e6fbe666649e16e91ce7a59d80b46e95d1841.jpg

* Economicus

Gpt-5.1

Deepseek-r1-0528

Gemini-2.5-pro

Qwen2.5-math-7b-instruct

Qwen3-30b-a3b-thinking-2507

* Olmo-3-32b-think

Olmo-3-7b-think

Qwen2.5-7b-instruct

Qwen3-30b-a3b-instruct-2507

- Olmo-3-7b-instruct

Gpt-4.1

Claude-haiku-4.5

Gemini-2.5-flash

Deepseek-chat

Human

• Economicus

Gpt-5.1

Deepseek-r1-0528

Gemini-2.5-pro
Qwen2.5-math-7b-instruct
Qwen3-30b-a3b-thinking-2507

- Olmo-3-32b-think
Olmo-3-7b-think
Qwen2.5-7b-instruct
Qwen3-30b-a3b-instruct-2507
- Olmo-3-7b-instruct
Gpt-4.1
Claude-haiku-4.5
Gemini-2.5-flash
Deepseek-chat
Human

The addition of open models also serves to pro-

开放模型的添加也起到了促进的作用

vide a deeper insight into what features of the training paradigm lead to the CM-RM distinction. The key, it appears, is explicit mathematical reasoning: models that are designed and trained for mathematical reasoning are consistently in the RM group, while those which are not fall into the CM group.

提供对训练范式中哪些特性导致了 CM-RM 区别的更深入理解。看起来关键在于显式的数学推理：那些被设计和训练用于数学推理的模型始终属于 RM 组，而那些没有被这样设计和训练的模型则属于 CM 组。

Moreover, the open models also allow us to observe the impact that model size has. Specifically, we find that in general, larger models tend to be more human-like and more economicus-like compared to their smaller counterparts. This can be viewed as a form of Pareto dominance, treating the Pareto frontier (the line in Figure 2) as providing the best tradeoffs in the human-to-economicus similarity space. Thus, smaller models are typically Pareto dominated by larger models. Moreover, an open, possibly smaller RM, can dominate a frontier CM (Qwen3-30B-thinking v.s. Gemini-2.5-Flash), suggesting that the size and training techniques both play a significant role. We do find that all frontier RMs are at or near the Pareto frontier.

此外，开放模型还使我们能够观察到模型规模的影响。具体来说，我们发现一般来说，较大的模型在人类相似性和经济人相似性方面，通常比较小的模型更加接近。这可以被视为一种帕累托优势，将帕累托前沿（图 2 中的线）视为在人类与经济人相似性空间中提供最佳权衡的边界。因此，较小的模型通常会被较大的模型所帕累托支配。此外，一个开放的、可能较小的 RM（如 Qwen3-30B-thinking）可以超越前沿的 CM（如 Gemini-2.5-Flash），这表明模型规模和训练技术在其中都起到了重要作用。我们确实发现所有前沿的 RM 都位于或接近帕累托前沿。

In our final aggregate analysis, we consider the decisiveness and consistency properties (i.e., relative invariance of decisions as we change the order and framing of prospects, as well as if we request an explanation) of LLMs, also comparing with the reference provided by the pool of human subjects.

在我们的最终综合分析中，我们考虑了大型语言模型（LLMs）的决断性和一致性属性（即，当我们改变前景的顺序和表述方式，以及请求解释时，决策的相对不变性），并将这些属性与由被试群体提供的参考进行比较。

Figure 2: Pairwise correlation between each frontier model and both the human and economicus references, shown as a 2D plot. Circles are CMs while x's are RMs.

图 2：每个前沿模型与人类和经济人参考之间的两两相关性，以二维图的形式展示。圆圈表示 CMs，而 x 表示 RMs。

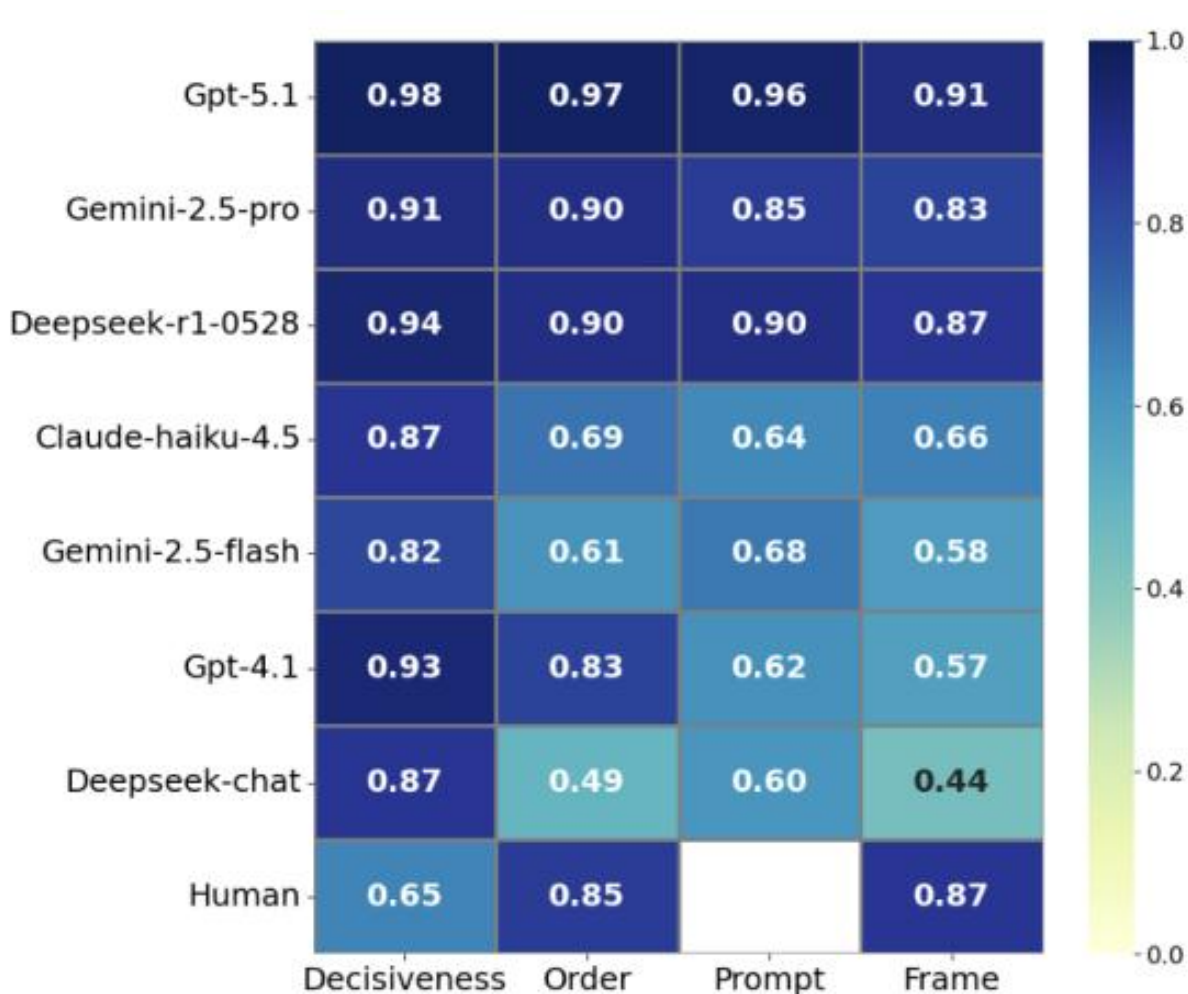


Figure 3: images/afe9281ee00f2b265936f2d354189b5c232ab973ccc2017bb7bde00e5290bb59.jpg

Figure 3: Consistency and decisiveness heatmap for frontier models and the human subjects.

图 3：前沿模型和人类受试者的一致性^①与决断性热力图。

The results provided in Figure 3 for the frontier models again exhibit a clear distinction between RMs and CMs along each of these dimensions. Actually, all LLMs appear to be rather decisive, especially as compared to the general human subjects pool, but RMs are nevertheless generally more decisive than CMs. The difference in terms of consistency is even more substantial. RMs are nearly order-invariant (just like economicus would be),

图 3 中提供的结果再次显示，在这些维度上，RM 和 CM 之间存在明显的区分。实际上，所有 LLMs 似乎都相当果断，尤其是与一般的人类受试者群体相比，但 RM 在总体上仍然比 CM 更加果断。在一致性方面的差异甚至更为显著。RM 几乎具有顺序不变性（就像经济人一样）。

whereas CMs tend to be strongly influenced by the order in which the prospects appear. CMs are also highly sensitive to the framing (gain vs. loss) and prompt (nature of explanation requested; more on this in Section 5), while RMs are not. Remarkably, both in terms of order and frame sensitivity, humans are far more like RMs. For example, surprisingly, human behavior appears to be considerably less influenced by gain/loss framing than CMs.

尽管 CMs 往往受到潜在客户出现顺序的强烈影响。CMs 还对框架（收益与损失）和提示（要求解释的性质；更多内容请参见第 5 节）高度敏感，而 RM 则不敏感。值得注意的是，在顺序和框架敏感性方面，人类与 RM 更为相似。例如，令人惊讶的是，人类的行为似乎比 CMs 受到收益/损失框架的影响要小得多。

4 Description-History Gap in LLMs

4 描述-历史差距在大语言模型中

In this section, we explore the description-history gap (DH gap)—that is, the behavior discrepancies between explicit (description) and implicit prospects (experience history)—of LLMs and human subjects in the same contexts x .

在本节中，我们探讨描述-历史差距（DH gap）——即在相同情境 x 下，大型语言模型（LLMs）和人类受试者在显性（描述）与隐性（经验历史）预期行为上的差异。

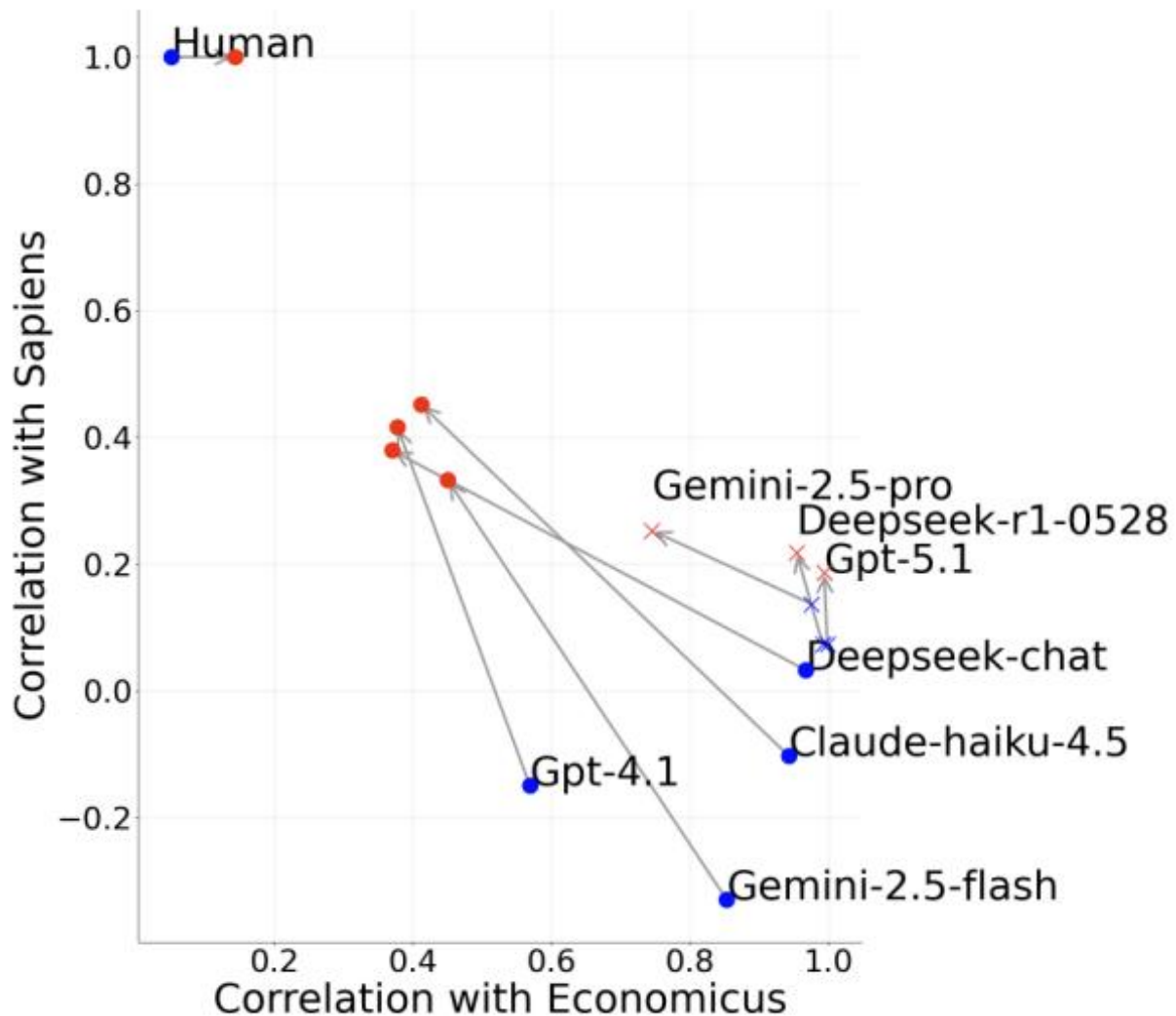


Figure 4: images/9770245ef9973e1925e2804c75c2a860de0aed260f54820b99cb69d0600dd5f6.jpg

Figure 4: HE representation of frontier LLMs (with human subjects as the reference). Change from explicit (blue) to implicit (red) prospects for each model is indicated by the arrows. RMs are x's and CMs are circles.

图 4：前沿大语言模型的 HE 表示（以人类受试者为参考）。箭头表示每个模型从显性（蓝色）到隐性（红色）前景的变化。RM 用 “x” 表示，CM 用圆圈表示。

The results, presented as the HE representation of each model as well as the human subjects, are provided in Figure 4 for the frontier LLM models and in Figure 5 for open LLMs. Consider first

the frontier LLMs, for which the observations are more crisp (Figure 4). An immediate and rather striking observation is that for all models—whether RMs (x's) or CMs (circles)—the change from explicit to implicit prospects increases the similarity to human behavior, and lowers similarity to economicus. An even more striking observation is the difference in how much the change to implicit prospects impacts RMs and CMs. In the case of the former, the impact tends to be moderate. For example, neither GPT-5.1 nor DeepSeek-R1 become especially less economicus-like when deciding from experience, though the change is more notable with Gemini Pro. For CMs, however, the change is dramatic: they become significantly more human-like, and significantly less economicus-like. Indeed, while RMs remain similar to one another in either setting, CMs occupy a relatively broad HE representation range under explicit prospects, but cluster closely with implicit prospects. For reference, we also show the change for the human subjects, who, somewhat surprisingly, become more economicus-like when prospects are presented implicitly; this likely accounts for the increased LLM-to-human correlation for RMs with implicit prospects.

结果以每个模型以及人类主体的 HE 表示形式呈现，如图 4 所示，适用于前沿的 LLM 模型；如图 5 所示，适用于开放的 LLM 模型。首先考虑前沿的 LLM 模型，这些模型的观察结果更为清晰（图 4）。一个直接且颇为显著的观察是，对于所有模型——无论是 RM（用 x 表示）还是 CM（用圆圈表示）——从显性预期转向隐性预期会增加与人类行为的相似性，并降低与 economicus 的相似性。更为显著的观察是，这种转变对 RM 和 CM 的影响程度存在差异。对于前者而言，这种影响往往较为温和。例如，当基于经验进行决策时，GPT-5.1 和 DeepSeek-R1 并不会变得明显不具有 economicus 特征，尽管在 Gemini Pro 中这种变化更为显著。然而，对于 CM 而言，这种转变则非常显著：它们变得明显更像人类，并且明显不那么像 economicus。事实上，虽然在两种情况下 RM 彼此之间的相似性保持不变，但 CM 在显性预期下占据了一个相对较广的 HE 表示范围，而在隐性预期下则紧密聚集在一起。作为参考，我们也展示了人类主体的变化情况，令人惊讶的是，当预期以隐性方式呈现时，人类主体变得更加像 economicus；这很可能是导致在隐性预期下 RM 与人类的相关性增加的原因。

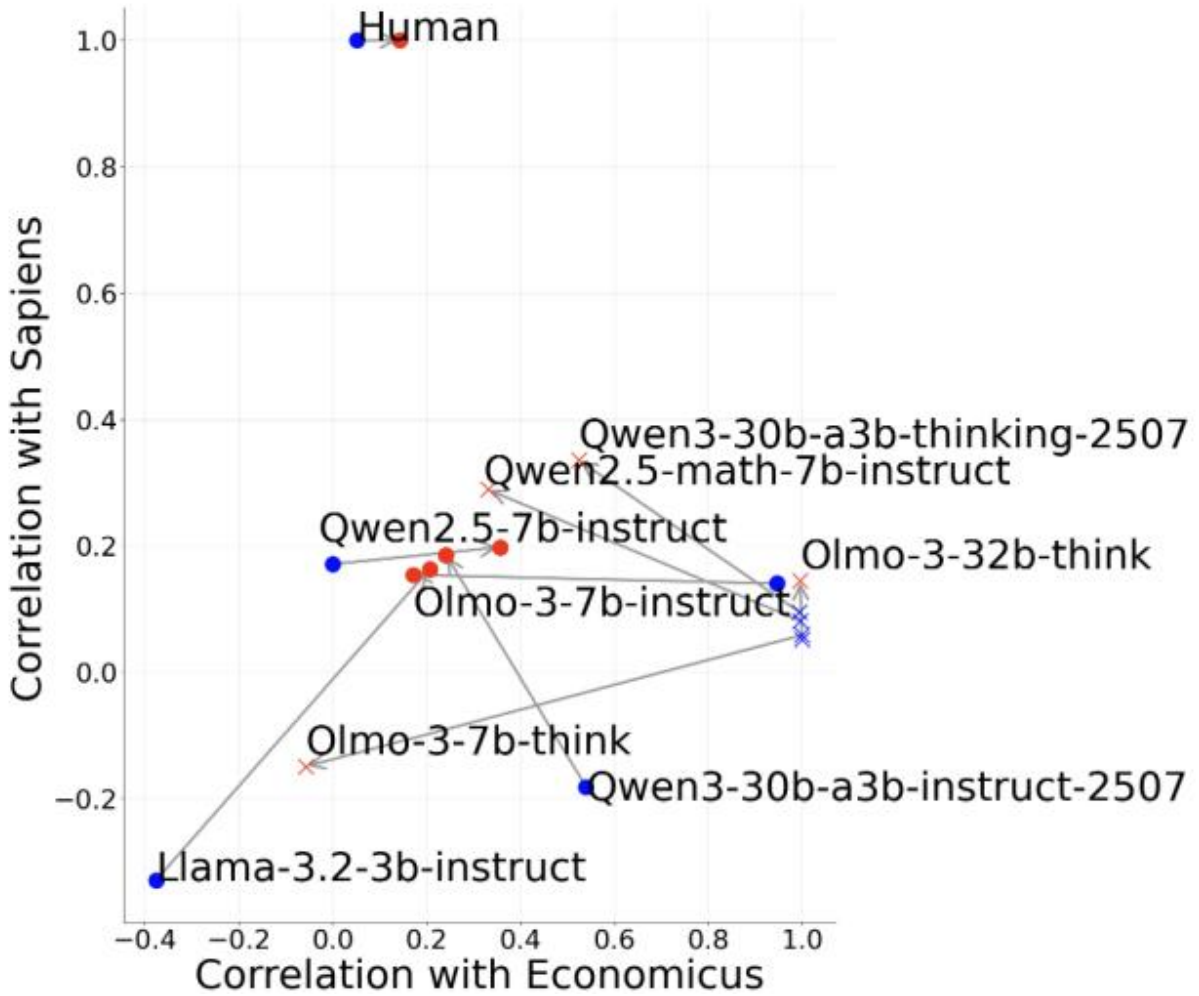


Figure 5: images/61c09d5ab1b6202f409df8a6dbc22940c4a310ef3e33826a2b651b1b0d640fe5.jpg

Figure 5: HE representation of open LLMs (with human subjects as the reference). Change from explicit (blue) to implicit (red) prospects for each model is indicated by the arrows. RMs are x's and CMs are circles.

图 5：开放大型语言模型的 HE 表示（以人类受试者为参考）。箭头表示每个模型从显性（蓝色）到隐性（红色）前景的变化。RM 用“x”表示，CM 用圆圈表示。

We can see some of the same trends with open LLMs (Figure 5), but with rather substantive differences. In particular, reasoning models no longer exhibit a small DH gap that we observed with frontier models: moving from explicit to implicit prospect representation, nearly all gaps are rather dramatic, with the bulk of the shift away from being economicus-like (i.e., to the left), although with a slightly more human-like shift as well. In addition, we can observe a non-trivial model size effect: the shifts are sharper, and differences from economicus-like behavior milder, with larger (30B and 32B) models than with smaller (7B) models. Nevertheless, we generally still see the RM vs. CM distinction, with the former models consistently to the right (more economicus-like) of the latter.

我们可以看到一些与开放型大语言模型（图 5）相似的趋势，但存在相当大的差异。特别是，推理模型不再表现出我们之前在前沿模型中观察到的小的 DH 差距：从显性到隐性前景表示的转变中，几乎所有差距都变得非常显著，总体上偏离了类似经济人（即向左）的模式，尽管也表现出略微更接近人类的转变。此外，我们还可以观察到一个非微不足道的模型规模效应：在较大的模型（如 30B 和 32B）中，转变更为明显，而与经济人行为的差异则较轻微，相比之下，在较小的模型（如 7B）中则不明显。尽管如此，我们通

常仍然可以看到 RM 与 CM 之间的区别，前者模型始终位于后者模型的右侧（更接近经济人）。

We note that all the models we tested have very long context windows. For example, DeepSeek-Chat has a 128K context limit. Even a 100-sample history represents only a small fraction of this avail-

我们注意到，我们测试的所有模型都有非常长的上下文窗口。例如，DeepSeek-Chat 的上下文限制为 128K。即使一个 100 个样本的历史记录，也仅占这个可用容量的一小部分。

able context, making it unlikely that the observed gap stems from context length limitations. This is further supported by results in the Appendix C showing that many LLMs exhibit more rational behavior with longer (100-sample) histories than with shorter (20-sample) histories, providing strong evidence that context constraints are not the underlying cause of the DH gap.

在上下文能力方面，这使得观察到的差距不太可能源于上下文长度的限制。这一点在附录 C 中的结果得到了进一步支持，这些结果显示许多大语言模型在使用更长（100 个样本）的历史时表现出比使用较短（20 个样本）历史时更合理的行为，这有力地证明了上下文限制并不是 DH 差距的根本原因。

5 Explaining Decisions

5 解释决策

Besides asking LLMs to make autonomous decisions under uncertainty, it is quite natural to additionally request a rationale—an explanation—of their decisions. However, at least when it comes to human decisions, providing such a rationale may itself impact the decision. Are LLMs similarly impacted when asked to provide an explanation?

除了要求大语言模型（LLMs）在不确定性下自主决策之外，要求它们提供决策的理由（即解释）也是很自然的。然而，至少在涉及人类决策时，提供这样的理由本身可能会对决策产生影响。当要求 LLMs 提供解释时，它们是否也会受到类似的影响？

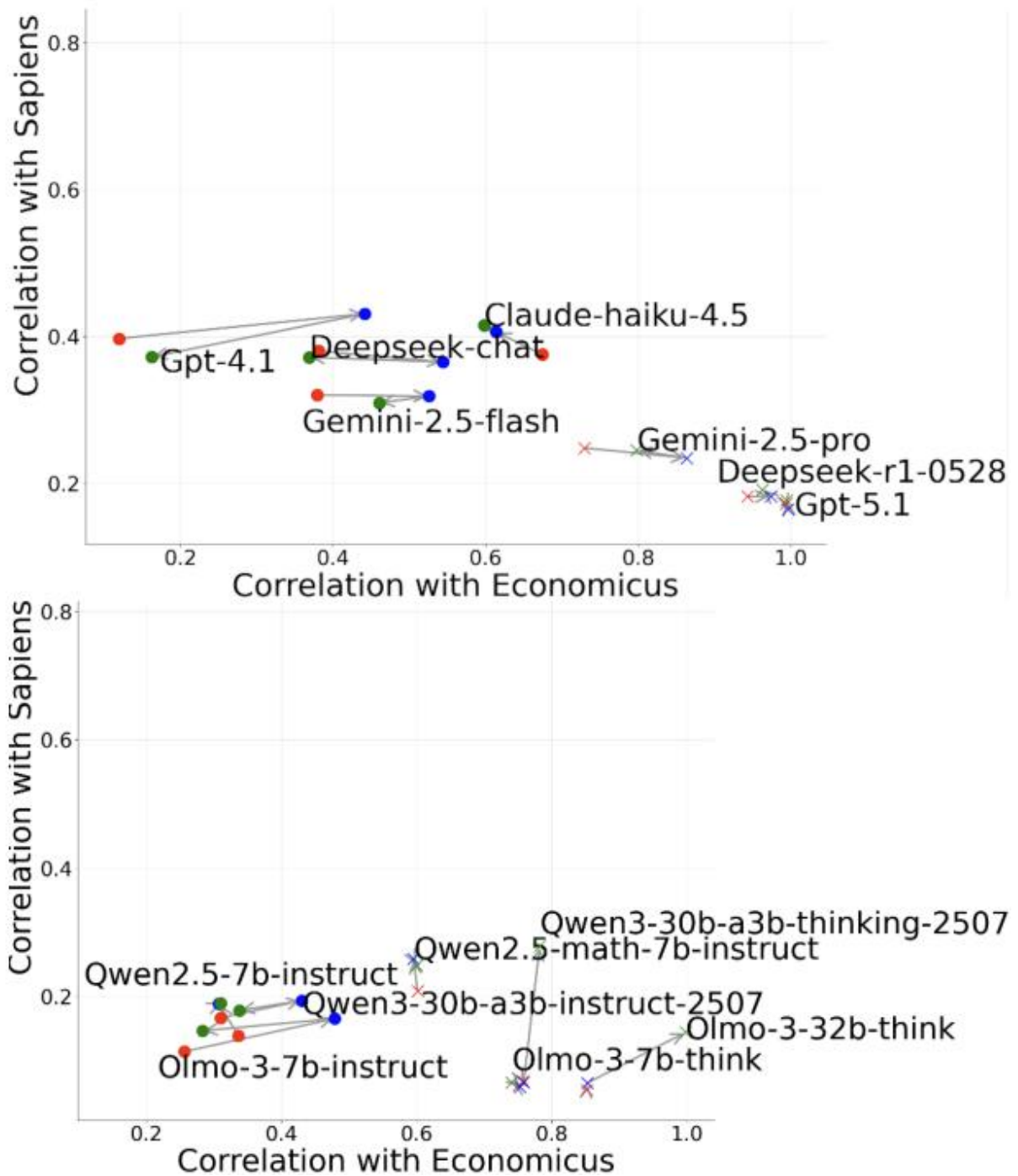


Figure 6: HE representation and the impact of explanations. Red: no explanation; Blue: one-sentence explanation; Green: math explanation.

图 6: HE 表示和解释的影响。红色：无解释；蓝色：一句解释；绿色：数学解释。

We investigate this by prompting LLMs for two kinds of explanations: a short (one-sentence) explanation and a full mathematical reasoning trace. Figure 6 visualizes the impact of these two explanation modes. What emerges is a surprising pattern, captured by the sequence of arrows starting at “no explanation” (red), then pointing to “short explanation” (blue), and then to “math explanation” (green). In particular, short explanations increase the degree of rationality (i.e., correlation with economicus) of most models, while asking

我们通过提示大型语言模型（LLMs）提供两种类型的解释来研究这一问题：一种是简短的（一句话）解释，另一种是完整的数学推理过程。图 6 可视化了这两种解释模式的影响。出现了一个令人惊讶的模式，

这一模式通过从“无解释”（红色）开始的一系列箭头来体现，然后指向“简短解释”（蓝色），再指向“数学解释”（绿色）。特别是，简短的解释提高了大多数模型的理性程度（即与经济人模型的相关性），而询问

for a longer rationale often reduces decision rationality in comparison with the former. Notably, this pattern is largely consistent for both frontier and open CMs, with Claude-4.5-Haiku somewhat of an outlier, whose both types of explanations result in behavior that is less like economicus compared to no explanation provided. Trends in RM behavior across explanation modes are more heterogeneous: More than half of the models (GPT-5.1, DeepSeek-R1, Olmo3-7B-think, and Qwen2.5-Math-7B-Instruct) exhibit strong stability; Gemini-2.5-Pro follows a pattern similar to the CMs; In contrast, Qwen3-30B-Thinking and Olmo3-32B-Think show significant changes when prompted to provide a mathematical explanation, while no differences are observed between the no-explanation and brief-explanation modes. One possible explanation is that, because CMs are not trained for mathematical reasoning, prompting them to provide mathematical explanations may only induce verbosity but no improved rationality.

对于更长的推理过程，通常会降低决策的理性程度，与前者相比。值得注意的是，这种模式在前沿型和开放型 CMs 中大致保持一致，而 Claude-4.5-Haiku 则是一个例外，其两种类型的解释所产生的行为与未提供解释的情况相比，更不像经济人（economicus）。在不同解释模式下，RM 行为的趋势更加多样化：超过一半的模型（如 GPT-5.1、DeepSeek-R1、Olmo3-7B-think 和 Qwen2.5-Math-7B-Instruct）表现出较强的稳定性；Gemini-2.5-Pro 的行为模式与 CMs 相似；相比之下，当被提示提供数学解释时，Qwen3-30B-Thinking 和 Olmo3-32B-Think 表现出显著的变化，而在未提供解释和简要解释模式之间则未观察到差异。一个可能的解释是，由于 CMs 并未接受数学推理的训练，提示它们提供数学解释可能只会导致冗长的输出，而不会提高其理性程度。

6 Interpretable Models of LLM Behavior

6 可解释的 LLM 行为模型

6.1 Analysis of Frontier Models

6.1 前沿模型分析

Our analysis thus far has considered the relationships among LLMs to one another, as well as to the behavior of human subjects and a rational risk-neutral utility maximizer (economicus). However, such comparisons yield only relatively coarse insight into actual behavioral tendencies. To dig deeper, we fit simple 4-parameter models of behavior to our data (see Section 2.3) for both LLMs and human subjects to offer interpretable characterizations of the underlying tendencies of LLMs that drive their decisions, contrasting these with both human subjects and economicus. Here, we present the results for the generalized power value model; results for the alternative parameterization are provided in Appendix C.7.

到目前为止，我们的分析已经考虑了大语言模型（LLMs）之间的关系，以及它们与人类被试和理性风险中性效用最大化者（economicus）的行为之间的关系。然而，这样的比较只能提供对实际行为倾向相对粗略的见解。为了更深入地探讨，我们为 LLMs 和人类被试拟合了简单的四参数行为模型（参见第 2.3 节），以提供对驱动 LLMs 决策的底层倾向的可解释性描述，并将这些描述与人类被试和 economicus 进行对比。在这里，我们展示了广义幂值模型的结果；替代参数化的结果见附录 C.7。

Model	σ	γ	β_{gain}	β_{loss}	Corr	MSE
DeepSeek-R1	0.86	0.81	832	711	0.98	0.007
Gemini-Pro	0.81	0.84	815	857	0.96	0.014
GPT-5.1	0.87	0.81	1000	802	1.00	0.001
Claude Haiku	0.89	0.80	221	244	0.71	0.053
DeepSeek-Chat	0.86	0.82	181	149	0.77	0.040

Gemini-Flash	0.93	0.89	27	42	0.87	0.015
GPT-4.1	0.79	2.17	205	0.01	0.55	0.050
Human	0.87	0.79	96	0.01	0.79	0.004

模型	σ	γ	β_{gain}	β_{loss}	纠正	MSE
DeepSeek-R1	0.86	0.81	832	711	0.98	0.007
Gemini-Pro	0.81	0.84	815	857	0.96	0.014
GPT-5.1	0.87	0.81	1000	802	1.00	0.001
Claude Haiku	0.89	0.80	221	244	0.71	0.053
DeepSeek-Chat	0.86	0.82	181	149	0.77	0.040
Gemini-Flash	0.93	0.89	27	42	0.87	0.015
GPT-4.1	0.79	2.17	205	0.01	0.55	0.050
人类	0.87	0.79	96	0.01	0.79	0.004

Table 1: Dual-Beta prospect theory parameters for decisions with explicit prospects.

表 1: 明确前景下的双 Beta 前景理论参数。

The results for explicit prospects are provided

显性前景的结果已提供

in Table 1 for the frontier models (corresponding confidence intervals are provided in Appendix C.5). First, we observe that human σ and γ are comparable to those of most LLMs, suggesting similar risk preferences and probability weighting across agents. The key distinction lies in decisiveness: humans exhibit moderate β_{gain} but near-zero β_{loss} , indicating they are far less consistent in the loss domain than in the gain domain. We can also note that human behavior appears to be relatively predictable by the simple 4-parameter prospect theoretic model, with $\text{MSE} < 0.01$ and correlation between the model and actual human behavior > 0.75 .

在表 1 中，关于前沿模型（相应的置信区间见附录 C.5）。首先，我们观察到人类 σ 和 γ 与大多数 LLMs 相当，表明代理之间的风险偏好和概率权重相似。关键的区别在于决策果断性：人类表现出适度的 β_{gain} 但接近零的 β_{loss} ，这表明他们在损失领域中的不一致性远大于收益领域。我们还可以注意到，人类行为似乎可以通过简单的 4 参数前景理论模型相对准确地预测，其均方误差（MSE）小于 0.01，模型与实际人类行为之间的相关性大于 0.75。

As we would expect, the frontier RMs exhibit behavior that is relatively close to rational (economicus): σ and γ are relatively close to 1, while both β_{gain} and β_{loss} are high. RMs are also extremely predictable by simple parametric models, with nearly perfect correlation between predictions and behavior. Moreover, even CMs’ behavior in this setting appears to be close to economicus, with the substantive difference from RMs being considerably smaller values of β_{loss} and β_{gain} (lower level of determinism). For the most part, CMs, too, are relatively well modeled by the simple 4-parameter models, with the exception of GPT-4.1, which has a lower correlation between predictions and behavior and a near-zero β_{loss} — resembling humans in being essentially random in the loss domain, though with high β_{gain} in the gain domain.

正如我们所预期的，前沿的 RMs 表现出与理性（economicus）相对接近的行为： σ 和 γ 相对接近 1，而 β_{gain} 和 β_{loss} 都较高。RMs 的行为也可以通过简单的参数模型进行高度预测，预测与行为之间的相关性几乎完美。此外，即使在这一设置中，CMs 的行为似乎也接近 economicus，与 RMs 的主要区别在于 β_{loss} 和 β_{gain} 的值显著较低（即确定性水平较低）。总体而言，CMs 也可以通过简单的四参数模型进行较好建

模，除了 GPT-4.1，其预测与行为之间的相关性较低，并且 β_{loss} 接近零——这与人类在损失领域表现出本质上随机的行为相似，但在收益领域则具有较高的 β_{gain} 。

Model	σ	γ	β_{gain}	β_{loss}	Corr	MSE
DeepSeek-R1	0.99	1.00	1000	346	0.98	0.009
Gemini-Pro	0.98	0.99	147	100	0.94	0.025
GPT-5.1	1.01	0.99	460	1000	0.98	0.010
Claude Haiku	1.07	0.98	4.81	23	0.55	0.078
DeepSeek-Chat	0.83	0.72	11	0.01	0.45	0.048
Gemini-Flash	1.57	0.82	2.61	14	0.58	0.047
GPT-4.1	1.53	0.74	0.01	39	0.58	0.106
Human	0.98	0.92	7	7	0.53	0.022

模型	σ	γ	β_{gain}	β_{loss}	纠正	MSE
DeepSeek-R1	0.99	1.00	1000	346	0.98	0.009
Gemini-Pro	0.98	0.99	147	100	0.94	0.025
GPT-5.1	1.01	0.99	460	1000	0.98	0.010
Claude Haiku	1.07	0.98	4.81	23	0.55	0.078
DeepSeek-Chat	0.83	0.72	11	0.01	0.45	0.048
Gemini-Flash	1.57	0.82	2.61	14	0.58	0.047
GPT-4.1	1.53	0.74	0.01	39	0.58	0.106
人类	0.98	0.92	7	7	0.53	0.022

Table 2: Dual-Beta prospect theory parameters for decisions with implicit prospects.

表 2: 隐含前景下的双 Beta 前景理论参数

Turning next to implicit prospects (Table 2), our first notable observation is that human behavior is now much closer to economicus: σ and γ are now all relatively close to 1, and notably β_{gain} and β_{loss} are now roughly equal, indicating that the strong gain-loss asymmetry in decisiveness observed with explicit prospects disappears. RMs largely remain close to economicus in this setting. Moreover, all three RMs remain highly predictable with the simple 4-parameter models: correlation is near-perfect, while MSE is quite low.

接下来转向隐含前景（表 2），我们的第一个显著观察是，人类行为现在与经济人（economicus）更为接近： σ 和 γ 现在都相对接近 1，值得注意的是 β_{gain} 和 β_{loss} 现在大致相等，这表明在显性前景中观察到的决策果断性中的强烈得失不对称性消失了。在此情境下，RMs 大体上仍然接近经济人。此外，所有三个 RMs 在使用简单的四参数模型时仍然高度可预测：相关性接近完美，而均方误差（MSE）相当低。

In the case of CMs, implicit prospects induce considerable variation in behavior. Claude Haiku and Gemini-Flash both show β_{loss} several times larger than β_{gain} , indicating greater decisiveness in the loss domain, while GPT-4.1 shows an even more extreme version of this pattern ($\beta_{\text{loss}} = 39$ vs. $\beta_{\text{gain}} = 0.01$). Most CMs exhibit some degree of probability overweighting ($\gamma < 1$), with the exception of Claude Haiku ($\gamma \approx 1$). All CMs have decisiveness that is now comparable to that of the human subjects pool, but several orders of magnitude below RMs. Notably, frontier CMs in both explicit and implicit prospect settings are considerably less predictable with 4-parameter prospect theory models than RMs.

在 CMs (认知模型) 的情况下, 隐含前景理论会导致行为产生显著变化。Claude Haiku 和 Gemini-Flash 都表现出比 β_{gain} 大几个数量级的 β_{loss} , 这表明在损失领域表现出更强的果断性, 而 GPT-4.1 则表现出这种模式的更加极端的形式 ($\beta_{loss} = 39$ vs. $\beta_{gain} = 0.01$)。大多数 CMs 表现出一定程度的概率过度权重 ($\gamma < 1$), 唯独 Claude Haiku 除外 ($\gamma \approx 1$)。所有 CMs 的果断性现在与人类受试者群体相当, 但比 RM (理性模型) 低几个数量级。值得注意的是, 在显性与隐性前景设定下, 前沿 CMs 在 4 参数前景理论模型中的可预测性明显低于 RM。

We note that these results differ in several ways from the alternative PT parameterization (Appendix C.7). Most strikingly, the standard model fits extreme values of σ and γ for human subjects ($\sigma = 0.04$, $\gamma = 0.13$ for explicit prospects), corresponding to strong risk aversion and probability distortion that sharply distinguish humans from LLMs. Under the dual-beta specification, however, human σ and γ are comparable to those of most LLMs ($\sigma = 0.87$, $\gamma = 0.79$); the distinction shifts entirely to the decisiveness parameters. This discrepancy likely arises because λ and β are confounded in the loss domain (see Appendix B.2.1), causing the standard model to compensate through distorted value function parameters. The standard model also exhibits instability in λ and β across RMs (e.g., $\lambda = 1000$ paired with $\beta = 1000$ for several models), further illustrating the identifiability issue. Despite these differences, both parameterizations achieve similar goodness of fit and yield qualitatively consistent conclusions regarding the RM-CM distinction.

我们注意到这些结果在多个方面与替代的 PT 参数化方法 (附录 C.7) 有所不同。最显著的是, 标准模型在人类受试者 (对于显性前景, 为 $\sigma = 0.04$ 和 $\gamma = 0.13$) 中拟合了 σ 和 γ 的极端值, 这对应于强烈的风险规避和概率扭曲, 这些特征明显地区分了人类与大型语言模型 (LLMs)。然而, 在双 β 参数化设定下, 人类的 σ 和 γ 与大多数 LLMs (如 $\sigma = 0.87$ 和 $\gamma = 0.79$) 相当; 区分点完全转移到了决断参数上。这种差异很可能源于在损失领域中 λ 和 β 的混淆 (参见附录 B.2.1), 导致标准模型通过扭曲的价值函数参数进行补偿。此外, 标准模型在 λ 和 β 上在不同 RM (例如, 多个模型中 $\lambda = 1000$ 与 $\beta = 1000$ 配对) 中表现出不稳定性, 进一步说明了可识别性问题。尽管存在这些差异, 两种参数化方法在拟合优度上仍表现出相似性, 并且在 RM-CM 区别的定性结论上也保持一致。

6.2 Analysis of Open Models: the Impact of Training and Scale

6.2 开源模型分析：训练与规模的影响

Our final analysis makes use of open models to investigate the impact of training paradigms on the effective LLM behavior. Table 3 presents the parameters for Qwen and Olmo models of different sizes before and after reasoning training phases. Our results largely confirm that training for mathematical reasoning consistently improves model rationality (including decisiveness) with explicit prospects. In addition, it significantly increases model predictability (in the sense of goodness of fit for the 4-parameter model as reflected by MSE). Notably, Qwen2.5-7B-Instruct and Qwen3-30B-

我们的最终分析利用开放模型来研究训练范式对有效大语言模型行为的影响。表 3 展示了推理训练阶段前后不同规模的 Qwen 和 Olmo 模型的参数。我们的结果在很大程度上证实了数学推理训练能够持续提高模型的合理性 (包括决断性), 并具有明确的前景。此外, 它显著提高了模型的可预测性 (以 4 参数模型的拟合优度 (通过均方误差 MSE 反映) 来衡量)。值得注意的是, Qwen2.5-7B-Instruct 和 Qwen3-30B-

Instruct both exhibit near-zero β_{loss} alongside much higher β_{gain} , echoing the human pattern of loss-domain randomness observed in Table 1.

指示两者均表现出接近零的 β_{loss} 以及显著更高的 β_{gain} , 这反映了表 1 中观察到的人类损失域随机性模式。

Model	σ	γ	β_{gain}	β_{loss}	Corr	MSE
Qwen2.5-7B-Instruct	0.84	0.75	57	0.01	0.64	0.030
Qwen2.5-Math-7B	0.85	0.82	1000	403	1.00	0.001

Qwen3-30B-Instruct	0.94	0.85	56	0.01	0.57	0.041
Qwen3-30B-Thinking	0.87	0.81	726	1000	0.99	0.004
Olmo-3-7B-Instruct	0.88	0.81	215	121	0.73	0.041
Olmo-3-7B-Think	1.33	1.16	1000	1000	1.00	0.001
Olmo-3-32B-Think	1.33	1.16	1000	1000	1.00	0.000

模型	σ	γ	β_{gain}	β_{loss}	纠正	MSE
Qwen2.5-7B-Instruct	0.84	0.75	57	0.01	0.64	0.030
Qwen2.5-Math-7B	0.85	0.82	1000	403	1.00	0.001
Qwen3-30B-Instruct	0.94	0.85	56	0.01	0.57	0.041
Qwen3-30B-Thinking	0.87	0.81	726	1000	0.99	0.004
Olmo-3-7B-Instruct	0.88	0.81	215	121	0.73	0.041
Olmo-3-7B-Think	1.33	1.16	1000	1000	1.00	0.001
Olmo-3-32B-Think	1.33	1.16	1000	1000	1.00	0.000

Table 3: Dual-Beta prospect theory parameters comparing instruction-tuned and reasoning models with explicit prospects.

表 3: 双 Beta 前景理论参数比较: 指令调优模型与显式前景推理模型

Model	σ	γ	β_{gain}	β_{loss}	Corr	MSE
Qwen2.5-7B-Instruct	48	0.66	0.01	2.45	0.46	0.070
Qwen2.5-Math-7B	0.55	0.88	7	0.82	0.54	0.027
Qwen3-30B-Instruct	1.10	0.83	17	2.84	0.50	0.047
Qwen3-30B-Thinking	1.00	0.88	668	585	0.94	0.025
Olmo-3-7B-Instruct	1.25	0.99	4.80	1.29	0.44	0.020
Olmo-3-7B-Think	0.73	0.71	1000	466	1.00	0.001
Olmo-3-32B-Think	0.95	1.07	1000	235	0.93	0.032

模型	σ	γ	β_{gain}	β_{loss}	纠正	MSE
Qwen2.5-7B-Instruct	48	0.66	0.01	2.45	0.46	0.070
Qwen2.5-Math-7B	0.55	0.88	7	0.82	0.54	0.027
Qwen3-30B-Instruct	1.10	0.83	17	2.84	0.50	0.047
Qwen3-30B-Thinking	1.00	0.88	668	585	0.94	0.025
Olmo-3-7B-Instruct	1.25	0.99	4.80	1.29	0.44	0.020
Olmo-3-7B-Think	0.73	0.71	1000	466	1.00	0.001
Olmo-3-32B-Think	0.95	1.07	1000	235	0.93	0.032

Table 4: Prospect theory model parameters comparing instruction-tuned and reasoning models with implicit prospects.

表 4: 比较指令调优模型和推理模型在隐含前景下的前景理论模型参数。

Table 4 presents analogous results with implicit prospects. Here, we see a similar effect of reasoning-based training with several of the models, although it appears to be somewhat less consistent than with explicit prospects.

表 4 展示了具有隐式前景的类似结果。在这里，我们看到基于推理的训练在几种模型中产生了类似的效果，尽管其一致性似乎不如显式前景的情况那样明显。

In the Supplement (Appendix C.4), we report checkpoints from each post-training stage (SFT, DPO, and RLVF) for both the instruction and thinking variants of the open models. As we observe no systematic effects attributable to either DPO or RLVF training, and since Olmo3-7B-Think and Olmo3-7B-Instruct share the same base model, one hypothesis can be that the initial SFT stage is the primary source of divergence between a reasoning and a conversational variant of models.

在补充材料（附录 C.4）中，我们报告了每个训练后阶段（SFT、DPO 和 RLVF）的检查点，适用于开放模型的指令和思维变体。由于我们观察到 DPO 或 RLVF 训练没有系统性影响，并且 Olmo3-7B-Think 和 Olmo3-7B-Instruct 共享相同的基模型，因此可以提出一个假设，即初始的 SFT 阶段是导致推理型模型和对话型模型之间差异的主要来源。

7 Conclusion

7 结论

We study LLM economic behavior through a controlled comparison spanning a diverse model suite and a human baseline. Beyond the canonical context manipulations of option order and outcome framing, we test two levers central to language-mediated decision making—the representation of risky options as explicit prospects versus outcome histories, and explanation prompting—and find that both systematically shift model choices. However, none of the evaluated models are fully human-like

我们通过一个受控比较研究，探讨大型语言模型（LLM）的经济行为，比较对象包括多样化的模型套件和人类基线。除了传统的上下文操控方式，如选项顺序和结果呈现方式之外，我们还测试了两个与语言中介决策密切相关的杠杆——将风险选项表示为明确前景与结果历史，以及解释提示。我们发现，这两种方式都会系统性地改变模型的选择。然而，评估的模型中没有一个是完全像人类的。

in this one-shot risky-choice setting; instead, behavior exhibits a robust two-cluster structure, separating “reasoning” and “conversational” models. Reasoning models are more invariant to contextual perturbations and tend to converge toward an expected payoff-maximizing economic baseline, whereas conversational models remain more context-sensitive and, in some conditions, display more human-like deviations. Our results suggest that LLM economic behavior reflects both model family and the decision interface: how alternatives are represented and responses elicited.

在这个一次性风险选择情境中；相反，行为表现出一个稳健的双聚类结构，将“推理”模型和“对话”模型区分开来。推理模型对上下文扰动更为不变，并倾向于收敛到一个预期收益最大化的“经济人”基准，而对话模型则更加依赖上下文，在某些条件下表现出更多类似人类的偏差。我们的研究表明，大语言模型的经济行为反映了模型家族和决策界面：选择方案如何被表示以及如何引发响应。

8 Limitations

8 局限性

Our analysis exhibits several limitations. First, our analysis of explanation prompting is incomplete on the human side: we did not elicit all explanation modes from human participants, and

we also do not disentangle whether effects depend on the ordering of “decide” versus “explain” (e.g., explanation-before-choice vs choice-before-explanation). Second, in our implicit (sample-based) representation, we aggregate two finite sample lengths (20 and 100 outcomes); because these settings can yield noticeably different behavior, a more systematic treatment of sample size is an important direction for future work. Third, while we vary several factors, fully characterizing their interactions is challenging: joint effects can be non-additive and can induce aggregation artifacts (e.g., Simpson’s paradox), and studying them reliably would likely require substantially more stimuli and a broader set of decision problems than the limited prospect families considered here. Finally, our prospect-theoretic parameterization is intended as a descriptive summary rather than a definitive mechanism: the four-parameter model can suffer from identifiability and boundary-fitting issues in this setting, so fitted parameters should be interpreted cautiously and primarily through robust qualitative comparisons rather than as precise estimates.

我们的分析存在几个限制。首先，在人类方面，我们对解释提示的分析并不完整：我们没有从人类参与者那里引出所有解释模式，而且我们也未能区分效应是否依赖于“决定”与“解释”顺序（例如，先解释后决定与先决定后解释）。其次，在我们的隐式（基于样本的）表示中，我们汇总了两个有限的样本长度（20 和 100 个结果）；由于这些设置可能会产生明显不同的行为，对样本量进行更系统的处理是未来工作的重点方向。第三，尽管我们变化了几个因素，但完全描述它们的相互作用具有挑战性：联合效应可能是非加性的，并可能引发汇总的偏差（例如，辛普森悖论），要可靠地研究这些效应，可能需要大量的刺激物和更广泛的决策问题，而不仅仅是这里考虑的有限前景家族。最后，我们的前景理论参数化旨在作为描述性总结，而非确定性机制：在这个设置中，四参数模型可能会遇到可识别性和边界拟合问题，因此拟合参数应谨慎解释，主要通过稳健的定性比较，而不是作为精确的估计。

Despite these limitations, our work makes important progress on several fronts. The RM–CM clustering provides the first systematic behavioral taxonomy of LLMs, the DH gap offers a novel window into how LLMs represent uncertain information, and our analysis of training effects (mathematical reasoning as the primary differentiator) is one of the first to directly test competing explanations for LLM behavioral variation.

尽管存在这些限制，我们的工作多个方面取得了重要进展。RM–CM 聚类为 LLMs 提供了首个系统的行为分类学，DH 差距为理解 LLMs 如何表示不确定信息提供了一个全新的视角，而我们对训练效果的分析（数学推理作为主要区分因素）是首个直接检验 LLMs 行为差异竞争解释的研究之一。

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